Chapter 5
Evaluation of Uncertainty in Direct Measurements



5.1 Concept of Measurement Uncertainty

Direct measurements are obtained by directly reading a measurement instrument that measures the same quantity.

In a direct measurement, the result is obtained by comparing the value read by the measuring instrument with the desired quantity. The measurement is done directly, without the use of mathematical equations.

Examples of direct measurements:

- Temperature measurement with a glass liquid thermometer (Fig. 5.1);
- Length measurement with a metric tape;
- Mass measurement with a scale;
- Pressure measurement with a gauge;
- Measurement of a thickness gear with a Vernier caliper (Fig. 5.2).

A measurement's result will always have a doubt associated with it, which we consider measurement uncertainty. What is sought in a measurement with metrological reliability is to estimate the measurement's results and associated uncertainty in the most reliable way possible.

Measurement uncertainty will always exist and never be eliminated since, as we have previously presented, the true quantity value is also estimated.

It is possible, however, to define the limits within which the value of a measurement with a certain associated probability is found.

The **measurement uncertainty** is defined by VIM—2.26 as:

- *Non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used.*
- *NOTE 1 Measurement uncertainty includes components arising from systematic effects, such as components associated with corrections and the assigned quantity values of measurement standards, as well as the definitional uncertainty.*

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In metrology, we usually adopt a confidence level of 95.45 % probability (remember that, in a normal distribution, 95.45 % probability represents two standard deviations). Therefore, when we say that the laboratory temperature is (21.0 ± 0.5) °C, we are saying that the true value of ambient temperature in the laboratory has a 95.45 % probability of being understood in this interval.

The uncertainty of a measurement's result is usually influenced by various components, which can be grouped into two categories according to the method characteristics used to estimate their numerical values: (1) uncertainties Type A and (2) uncertainties Type B, which will be detailed below.

5.2 Types of Measurement Uncertainties

Various sources of uncertainty exist in a measurement. Therefore, we should estimate these uncertainties and minimize their influences so that the measurement outcome is known in a smaller interval.

5.2.1 Type A Evaluation of Measurement Uncertainty

According to VIM—2.28, it consists of evaluating *"a component of measurement uncertainty by a statistical analysis of measured quantity values obtained under defined measurement conditions."*

Type A uncertainties can, therefore, be characterized by experimental standard deviations. In metrology, the best estimate of a quantity that varies randomly is the arithmetic mean x of n measurements made. The estimated variance (s^2) or the estimated standard deviation (s) characterizes the variability of the measured values, that is, the dispersal around the mean value.

The best estimate of the variance of the mean is the experimental mean-variance $s^2(\bar{x})$, whose expression is:

$$s^2(\bar{x}) = \frac{s^2}{n} \tag{5.1}$$

The experimental standard deviation of the medium $\bar{x}$ serves to qualify how much the mean value $\bar{x}$ represents the quantity to be measured. The better this estimate, the greater the number of repetitions made in the measurement.

Important
The equation $s(\bar{x}) = \frac{s}{\sqrt{n}}$ determines Type A uncertainty measurement or the repeatability measurement uncertainty.

For several reasons, especially economic ones, the number of repetitions of a measurement is reduced, usually ranging from three to ten.

5.2.2 Type B Evaluation of Measurement Uncertainty

According to VIM—2.29:

Evaluation of a component of measurement uncertainty determined by means other than a Type A evaluation of measurement uncertainty.
EXAMPLES Evaluation based on information

- *associated with authoritative published quantity values,*
- *associated with the quantity value of a certified reference material,*
- *obtained from a calibration certificate,*
- *about drift,*
- *obtained from the accuracy class of a verified measuring instrument,*
- *obtained from limits deduced through personal experience.*

Type B uncertainties can be characterized by standard deviations estimated by distributions of probabilities assumed or based on experience or other observations. Accessory and external information to the measurement process—obtained from previous measurements of similar measurements, experience or knowledge of measuring instrument behavior, manufacturer data, data provided by calibration certificates, and instruction manual references—allows you to determine the uncertainties of this type.

Examples of Type B uncertainty:

- Temperature gradient during measurement;
- Difference of ambient temperature concerning the stipulated reference temperature;
- Indicator reading resolution;
- Stability of the power supply;
- Parallax error;
- Uncertainty of the measurement standard;
- Drift from the standard;
- Geometric errors;
- Mechanical deformations;
- Hysteresis error.

In the evaluation of Type B uncertainty, it is necessary to consider and include, when pertinent, at least those originating from the following sources:

(a) The uncertainty associated with the reference standard and any instability in its value or indication (standard subject to instrumental drift or temporal instability).

5.3 Evaluations of More Frequent Type B Uncertainties

As we have seen, the estimate of Type A uncertainty is obtained by calculating the standard deviation of the mean measurement. The estimate of Type B uncertainties already has several origins. Next, we present the primary sources of Type B uncertainty and how to calculate them.

5.3.1 Estimation of the Uncertainty of Reading Resolution

It is essential to evaluate the contribution of reading resolution in estimating measurement uncertainty, as it is widespread to find a low dispersion of the values obtained in a measurement process, which characterizes Type A uncertainty as being "zero." In this case, depending on the value of the resolution and the type of probability distribution adopted, this uncertainty may be one of the largest or the most significant contribution to final uncertainty.

In a measurement process, we can come across two situations.

Situation 1: Measurement where we "seek" the value of the desired quantity; that is, we do not know a priori what the value is.

Example 5.1 Reading Obtained ON A Digital Scale
Suppose the value of the mass of an object is 25.9 g and that the digital scale used for this measurement has a resolution of 0.1 g. This means that the lowest value read by the scale is 0.1 g.

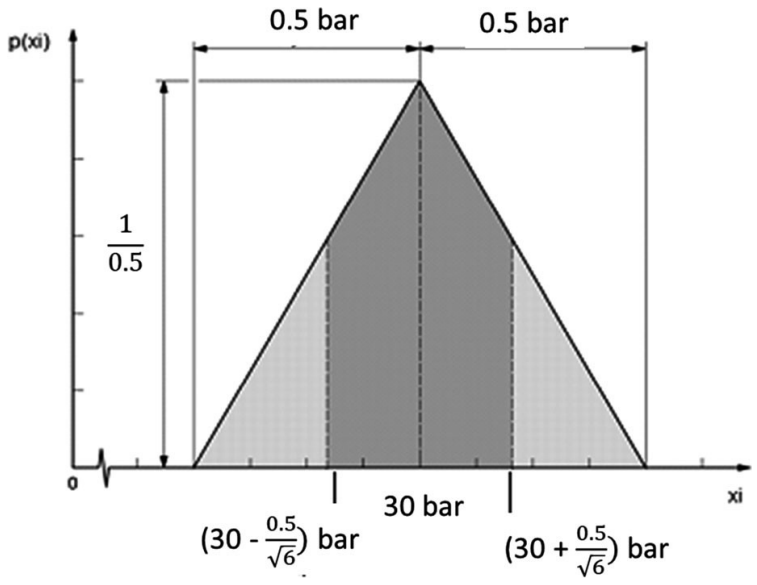
Considering the algorithm in the digital scale responsible for digitizing the values indicated, the "true value" of the mass will be comprised between the interval [25.85 and 25.949 …] g. Values such as 25.95 g or larger should be rounded by the instrument to 26.0 g, just as values such as 25.84 g or smaller to 25.8 g.

Therefore, every time the scale indicates 25.9 g, we will doubt the "true value" of the mass caused by its resolution limitation. Considering that the probability that the "true value" is understood between [25.85 and 25.949 …] g is the same within this interval, it is reasonable to adopt a statistical distribution that reflects this behavior, that is, rectangular or uniform distribution. Graph 5.1 shows:

Note that the uncertainty of reading resolution will be the standard deviation of the rectangular distribution, that is:

$$u_{res} = \frac{R}{\sqrt{12}} \tag{5.2}$$

Where R is the resolution adopted.



Graph 5.2 Uncertainty of pressure gauge resolution (triangular distribution)

Considering that the probability of "true value" is higher at point 30 bar than at any other point, because we fix this value, it is reasonable to adopt a statistical distribution that reflects this behavior, that is, the triangular distribution.

In Graph 5.2, we have:

The uncertainty of reading resolution will be the standard deviation of the triangular distribution with R = 1 bar.

$$u_{res} = \frac{R}{\sqrt{24}} = \frac{1}{\sqrt{24}} bar = 0.20\ bar \tag{5.3}$$

Given the limitations of the digitization algorithm, the digital instrument will adopt a uniform probability in rounding readings regardless of whether or not we fix the desired value.

For the analog instrument, the probability of the measurement may be triangular or uniform, depending on whether or not we fix the reading value.

5.3.2 Reading Resolution Adopted by the Calibration Laboratory

Let us look at the following situation: using a calibrated instrument to perform a measurement.

Some questions

1. Given that the laboratory that calibrated our instrument has incorporated the uncertainty of reading resolution in the estimate of final uncertainty, we must, in our measurements, consider the reading resolution of this calibrated standard as

one of the components, as "it is considered" in the uncertainty declared on the certificate?
2. When considering the resolution, are we not repeating the same source of uncertainty twice?
3. In the case of a digital standard, the value of the resolution adopted by the calibration laboratory is known as it will be equal to the digital increase.
4. In the case of an analog standard, a question arises: Was the value of reading resolution adopted by the laboratory the same as what we will adopt?
5. Is it possible to place a magnifying glass on the analog instrument reading scale, thus reducing its measurement uncertainty, since we are reducing its reading resolution as long as it does not make it lower than the instrument sensitivity? This is a usual procedure in laboratories, but the end user of the instrument should be informed.

Answering the questions.
If the user can repeat, in reading with the instrument that came from calibration, the resolution adopted by the laboratory during calibration, the contribution of the standard resolution should not be considered in the final uncertainty of the measurement. Otherwise, this portion should be considered in the final estimate.

For this reason, the calibration laboratories must provide the instrument's calibration certificate with the resolution adopted in calibration. Thus, we will know the value of the resolution adopted and can repeat it at the time of measurement with this instrument.

Using a magnifying glass to read analog instruments is allowed and healthy. However, we should not determine the resolution of a measuring instrument with a magnifying glass. It will lower the resolution than what can be discerned with the "naked eye," and the user usually reads with the "naked eye."

5.3.3 Hysteresis Uncertainty Estimate

In Chap. 4, we saw that hysteresis is the biggest difference between a measurement instrument's charge and discharge values. Scales, comparator clocks, gauges, among others, most commonly present hysteresis errors.

To estimate the uncertainty of hysteresis, we calculate the instrument's hysteresis (H) at the point and adopt a uniform or rectangular probability distribution.

$$u_{\text{hysteresis}} = \frac{H}{\sqrt{12}} \tag{5.4}$$

Solved Exercise 5.1.
A Bourdon-type gauge, whose measurement range is 0 to 20 kgf/cm^2, was calibrated by comparison with a standard gauge. The values found are in Table 5.1.

Knowing a Little More…
In calibrating the gauge, we fix the object's pointer directly to the desired pressure and verify the variation of its pressure using the standard pressure gauge. Thus, when we fix the pressure at 5 kgf/cm^2, for example, we realize the variation of the object gauge in the standard. This is why, at point 5 kgf/cm^2, we have the standard marking (5.0, 5.2, 5.0, and 5.2) kgf/cm^2. This variation is not caused by the standard but by the calibration object gauge.

5.3.4 Evaluation of the Uncertainty of the Standard Instrument

A source of Type B uncertainty always exists in the calibration of measurement instruments: the uncertainty from the standard instrument. In Chap. 2, when we define calibration, we highlight that calibrating confronts the values measured by the standard instrument with the calibration instrument (object). Therefore, the object instrument's uncertainty inherits the standard instrument's uncertainty.

To determine the uncertainty of the standard instrument, check this value in the standard instrument calibration certificate.

5.4 Standard Measurement Uncertainty

According to VIM—2.30, we have: *"measurement uncertainty expressed as a standard deviation."*

We must express all components of uncertainty (u_i) of Types A and B corresponding to a standard deviation. For this, we need to evaluate the probability distribution applied to the uncertainty: normal distribution, rectangular or uniform distribution, triangular distribution, etc.

5.5 Combined Standard Measurement Uncertainty

According to VIM—2.31, we have: *"standard measurement uncertainty that is obtained using the individual standard measurement uncertainties associated with the input quantities in a measurement model."*

The equation can briefly determine combined standard uncertainty (u_c):

$$u_c = \sqrt{u_A^2 + u_B^2} \tag{5.5}$$

Where u_A are all Type A uncertainties, and u_B are all Type B uncertainties.

5.6 Effective Degrees of Freedom

The definition of ISO GUM says the following for the degree of freedom: *"In general, the number of terms in a sum minus the number of constraints on the terms of the sum."*

When more than 30 measurements of the same measurement are performed, we know, through statistics, that these results are very close to a normal distribution; if a smaller number of measurements is used, we must bring this distribution closer to a normal distribution by applying the t-distribution correction factor. However, to establish this correction factor, it is necessary to determine the number of effective degrees of freedom of the distribution.

When various sources of uncertainty are considered to estimate combined standard uncertainty (u_C), the number of effective degrees of freedom resulting from combined uncertainty has to be calculated from information from each source of uncertainty. Therefore, it is recommended to use the Welch-Satterthwaite equation to estimate the number of effective degrees of freedom:

$$\frac{u_c^4}{\upsilon_{eff}} = \frac{u_1^4}{\upsilon_1} + \frac{u_2^4}{\upsilon_2} + \ldots + \frac{u_i^4}{\upsilon_i} \tag{5.6}$$

Where u_C is the combined standard measurement uncertainty; $u_1, u_2, \ldots, u_i$ are the standard measurement uncertainties of each source of uncertainty (Type A and Type B uncertainty); ν_1, ν_2, ν_3 …, υ_i are the degrees of freedom of each i source of uncertainty, and υ_{eff} is the number of effective degrees of freedom associated with combined standard uncertainty.

Eq. (5.6) can be reordered and presented by:

$$\upsilon_{eff} = \frac{u_c^4}{\sum \frac{u_i^4}{\upsilon_i}} \tag{5.7}$$

Important
The degree of freedom associated with the uncertainty of repeatability (Type A) equals $n - 1$, where n is the number of measurements.

(continued)

In evaluating the degree of freedom of Type B standard uncertainty from a probability distribution a priori, for example, a uniform or triangular distribution, it is implicitly supposed that the value of uncertainty resulting from such evaluation is known precisely. This implies that the degree of freedom associated with this uncertainty will be infinite.

5.7 Coverage Factor

VIM—2.38 defines it as the *"number larger than one by which a combined standard measurement uncertainty is multiplied to obtain an expanded measurement uncertainty. NOTE: A coverage factor is usually symbolized k"*.

The coverage factor $\boldsymbol{k}$ should always be declared so that the standard uncertainty of quantity measurement can be recovered to calculate the combined standard uncertainty of other measurement results that eventually depend on this quantity.

This factor k will be obtained from the determination of the number of effective degrees of freedom ($\boldsymbol{\nu_{eff}}$) and using the t-distribution, in which the value of t will be the coverage factor k.

The value of $\boldsymbol{\nu_{eff}}$ obtained by Eqs. (5.6) or (5.7) is usually not an integer. From the effective degree of freedom, the coverage factor k can be obtained from Excel, using the INV.T.BC function, or in the *Student's t-table*.

When we use the calculated value of the $\boldsymbol{\nu_{eff}}$ in the t-table, we should always approach it to the immediately lower integer. For example, if the computed value is $\boldsymbol{\nu_{eff}} = 10.46$, we must enter the table with $\boldsymbol{\nu_{eff}} = 10$ and obtain $k = 2.28$. This will be the value used for the coverage factor k.

Important
By using the calculated value of the effective degree of freedom ($\boldsymbol{\nu_{eff}}$) in the Student's t-table, approach it to the immediately lower integer. This will ensure a more prominent coverage factor and, thus, more significant expanded uncertainty.

5.8 Expanded Measurement Uncertainty

According to VIM—2.35, expanded measurement uncertainty is defined as:

Product of a combined standard measurement uncertainty and a factor larger than the number one.

NOTE 1 The factor depends upon the type of probability distribution of the output quantity in a measurement model and on the selected coverage probability.

NOTE 2 The term "factor" in this definition refers to a coverage factor.

NOTE 3 Expanded measurement uncertainty is termed "overall uncertainty" in paragraph 5 of Recommendation INC-1 (1980) (see the GUM) and simply "uncertainty" in IEC documents.

The expanded uncertainty $\boldsymbol{U}$ is the product of the combined standard uncertainty $\boldsymbol{u_C}$ and the coverage factor $\boldsymbol{k}$:

$$U = k.u_c \tag{5.8}$$

Multiplication of combined standard uncertainty by a constant does not provide additional information. It is just a way of representing the final uncertainty associated with a coverage probability. In calibration and industrial measurements, it is expected to adopt the probability of 95.45 %, which would correspond, in a normal distribution, to a coverage factor equal to two.

Important
We must always combine standard uncertainties with one standard deviation. Therefore, when we use the uncertainty of measuring from a calibration certificate, we must divide it by the scope factor k since the uncertainties declared in a calibration certificate are expanded to 95.45 %.

5.9 Presentation of the Measurement Result

According to the document ILAC-P14:09/2020—ILAC Policy for Measurement Uncertainty in Calibration, section 5.3: "*The numerical value of the expanded uncertainty shall be given to, at most,* ***two significant digits****. Where the measurement result has been rounded, that rounding shall be applied when all calculations have been completed; resultant values may then be rounded for presentation. For the process of rounding, the usual rules for rounding of numbers shall be used, subject to the guidance on rounding provided,* i.e.*, in Sect. 7 of the GUM*."

5.10 Sources of Measurement Uncertainty

Next, we will present some sources of frequent measurement uncertainties in various areas of metrology.

5.10.1 Dimensional Metrology

- Measurement uncertainty of the standard: this information is in the standard calibration certificate.
- Temperature effect: the temperature difference between measuring, the standard, and the temperature of the calibration laboratory should be considered. As a rule, the ambient temperature of the calibration laboratory should be 20.0 °C. This effect is more significant for considerable lengths and in cases where the measurand is of a material different from the standard. Although it is possible to correct these errors, residual uncertainties of the uncertainty of the coefficients of dilation and uncertainty in the calibration of the thermometer will always remain.
- Elastic deformation at the point of contact: is critical in the most exact measurements and in the cases involving different materials. Its magnitude is a function of the measurement force and the nature of the contact between driving and measurand. Although it is possible to correct the results of these errors, the uncertainty of this correction should be considered due to the uncertainty of the applied force and the physical properties of the components in contact.
- Cosine error: misalignment between the measurand or standard and the measurement axis. Residual errors will often persist by the assumption that reference surfaces are exempt from geometric errors.
- Geometric error: planning or spherical errors, parallelism or perpendicularity of the support surface, measurand, or standard cylindrical error.
- Doubt in reading: uncertainty in the resolution of the instrument.
- Stability of the standard, or measurand, as a function of time.

In this area, in general, the following instrument calibration intervals are used (Table 5.4):

5.10.2 Thermal Metrology

- Measurement uncertainty of the standard: this information is in the standard calibration certificate.
- Electrical equipment/instruments used as support: standard resistors uncertainty, multimeters, power supplies, thermal baths, etc.

Table 5.4 Calibration interval

Instrument	Calibration interval—months
Measuring tape	6
Vernier caliper	12
Micrometer	12
Planger dial gauge	12
Slip gauges (standard block)	12

Table 5.5 Calibration interval

Instrument	Calibration intervals—months
Glass thermometer	6 a 12
Resistance thermometer (Pt-100)	12
Thermocouple	12
Bimetallic thermometer	12

- Doubt in reading: uncertainty in the resolution of the instrument.
- Partial immersion in glass thermometers: the part of the column of the immersion thermometer outside the medium provides a difference in the temperature indication.
- Effect of resistance thermometers self-heating: the sensor is heated by the current circulating.
- Parasitic electrical uncertainties: uncertainties of electrical origin resulting from static electricity at contact terminals. We can estimate this $\frac{2\mu V}{\sqrt{3}}$ value primarily when we calibrate thermocouples.
- Drifts of standards and electrical instruments.

In this area, in general, the following instrument calibration intervals are used (Table 5.5):

5.10.3 Mass Metrology

- Measurement uncertainty of the mass standard: this information is in the standard calibration certificate.
- Drift of the masses as a function of time: change of standard mass measurement error as a function of time, depending on the surface finish and the quality of manufacture, material type, handling, atmospheric corrosion, etc. Without this information, we replace it with the maximum permissible error.
- Environmental conditions: temperature gradients, humidity, static electricity.
- Doubt in reading: uncertainty in the resolution of the instrument.
- Air buoyancy: the density of the air can be determined from the measurement of atmospheric pressure, temperature, and relative humidity. Even when the density is corrected, uncertainties of pressure, temperature, and humidity measurements will be present.
- Measurement process: the quality of the scale influences the result of the measurement and, therefore, we must know its characteristics:
  - Repeatability of measurements;
  - Linearity;
  - Eccentricity of the load, especially when more than one mass is placed on the plate;
  - Influence of magnetic fields;

Table 5.6 Calibration interval

Instrument	Calibration interval—months
Standard mass	24 to 48
Precision scales	12 to 36
Analytical scales	12

Table 5.7 Calibration interval

Instrument	Calibration interval—months
Digital multimeter	12
Oscilloscope	12 to 36
Resistive decade	24 to 48

- Effects of temperature;
- Lever arm's length.

In this area, in general, a calibration interval of the instruments is used as follows (Table 5.6):

5.10.4 Electric Metrology

- Uncertainty of electrical reference standards: this information is in the standard calibration certificate.
- Different environmental conditions from the recommended.
- Measurement system stability: as a function of time and conditions of use.
- Doubt in reading: uncertainty in the resolution of the instrument.
- Impedance of cables, terminals, and instruments: parasitic electrical uncertainties resulting from static electricity at contact terminals. Its value is estimated in $\frac{2\mu V}{\sqrt{3}}$.
- Layout of instruments and standards during calibration: current leaks, electromagnetic fields, grounding.

In this area, in general, a calibration interval of the instruments is used as follows (Table 5.7):

5.10.5 Pressure Metrology

- Measurement uncertainty of the standard: This information is in the standard calibration certificate.
- Drift of the masses as a function of time: change of standard mass measurement error as a function of time, depending on the surface finish and the quality of manufacture, material type, handling, atmospheric corrosion, etc. Without this information, we replace it with the maximum permissible error.
- Doubt in reading: uncertainty in the resolution of the instrument.

- Hysteresis.
- Different environmental conditions from the recommended.
- Drifts of standards and electrical instruments.
- Impedance of cables, terminals, and instruments: parasitic electrical uncertainties resulting from static electricity at contact terminals. Its value is estimated in $\frac{2\mu V}{\sqrt{3}}$.

5.10.6 Analytical Metrology

- *Sampling: where in-house or field sampling forms part of the specified procedure, and effects such as random variations between different samples and any potential for bias in the sampling procedure form components of uncertainty affecting the final result.*
- *Storage Conditions: where test items are stored for any period prior to analysis, the storage conditions may affect the results. The duration of storage, as well as conditions during storage, should, therefore, be considered uncertainty sources.*
- *Instrument effects: may include, for example, the limits of accuracy on the calibration of an analytical balance; a temperature controller that may maintain a mean temperature that differs (within specification) from its indicated set point; an auto-analyzer that could be subject to carryover effects.*
- *Reagent purity: the concentration of a volumetric solution will not be known precisely even if the parent material has been assayed, since some uncertainty related to the assaying procedure remains. Many organic dyestuffs, for instance, are not 100 % pure and can contain isomers and inorganic salts. The purity of such substances is usually stated by manufacturers as being not less than a specified level. Any assumptions about the degree of purity will introduce an element of uncertainty.*
- *Assumed stoichiometry: where an analytical process is assumed to follow a particular reaction stoichiometry, it may be necessary to allow for departures from the expected stoichiometry, or incomplete reaction or side reactions.*
- *Measurement conditions: for example, volumetric glassware may be used at an ambient temperature different from that at which it was calibrated. Gross temperature effects should be corrected, but any uncertainty in the temperature of liquid and glass should be considered. Similarly, humidity may be important where materials are sensitive to possible changes in humidity.*
- *Sample effects: the recovery of an analyte from a complex matrix, or an instrument response, may be affected by composition of the matrix. Analyte speciation may further compound this effect. The stability of a sample/analyte may change during analysis because of a changing thermal regime or photolytic effect. When a 'spike' is used to estimate recovery, the recovery of the analyte from the sample may differ from the recovery of the spike, introducing an uncertainty that needs to be evaluated.*

measurement is stable with very low or no variation in instrument resolution, we can proceed according to the following exercise.

Exercise: A glass liquid thermometer measures fuel oil temperature with a measurement uncertainty of 0.1 °C (for k = 2 and 95.45 %) and instrumental bias of − 0.2 °C. The value found was 28.4 °C. Determine the measurement result.
Solution: Corrected measurement = (28.4 + 0.2) °C = 28.6 °C

$$MR = (28.6 \pm 0.1) \ ^\circ C$$

Note that it was impossible to calculate the uncertainty of the repeatability of the measurement. Thus, the declared measurement instrument in its calibration certificate only inherited the final uncertainty.

5.11 Proposed Exercises

5.11.1 A car speedometer ranges from (0 to 200) km/h. The uncertainty at any point is 2 km/h

(a) What is the uncertainty at 100 km/h?
(b) What is the percentage uncertainty compared to 100 km/h?
(c) What is the percentage uncertainty to 50 km/h?
(d) What is the percentage uncertainty compared to 5 km/h?
(e) At what point is the lowest percentage uncertainty?

5.11.2 Mike measures his brother's height and finds 176.35 cm, with an uncertainty of 0.21 cm.

(a) Round out and write the height of Mike's brother with one significant digit concerning his uncertainty.
(b) Give the same answer in meters.

5.11.3 Martha uses a timer to measure the period of a pendulum. The results are (Table 5.9):

(a) What is the mean value of the period?
(b) What is the standard deviation of the mean?

Table 5.9 Period measurements

Measurements	Period (s)
1	0.63
2	0.64
3	0.65
4	0.63
5	0.65